

Notes: Ivković and Weisbenner (JF, 2005)
**Local Does as Local Is: Information Content of the Geography of Individual
Investors' Common Stock Investments**

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1 Big picture

- **Research question.** Do individual investors' local stockholdings reflect value-relevant local information, or do households simply buy nearby firms because they are familiar with them?
- **Main idea.** If local bias is only familiarity, local holdings should not systematically outperform nonlocal holdings. If local bias reflects information, local stocks in household portfolios should outperform comparable nonlocal stocks, especially for firms with greater information asymmetry.
- **Setting.** The paper uses detailed portfolio, trade, and location data for retail investors at a large discount broker from 1991 to 1996. It links household zip codes to firm headquarters and classifies a stock as local if the firm's headquarters are within 250 miles of the household.
- **Local bias.** Households invest much more locally than a market benchmark would imply. At the end of 1991, 31.5% of the average household stock portfolio is local, compared with only 12.6% of all firms being within 250 miles.
- **Main performance result.** Local holdings outperform nonlocal holdings by about 3.2% over the next year in the raw-return specification. With zip-code fixed effects, the one-year gap is 4.2%.
- **Information-asymmetry result.** The return premium is concentrated in non-S&P 500 stocks. Local non-S&P 500 holdings beat nonlocal non-S&P 500 holdings by 6.2% over the next year and by 20.9 basis points per month in four-factor alpha.
- **Portfolio-mimicking result.** A zero-cost portfolio that is long local holdings and short nonlocal holdings earns positive returns. The effect is strongest for non-S&P 500 stocks: roughly 25 basis points per month in raw returns and 20 basis points per month in four-factor alpha.
- **Local ownership signal.** Nonlocal non-S&P 500 stocks with high relative local ownership perform much better than nonlocal non-S&P 500 stocks with low relative local ownership. In the authors' phrase, if one cannot be local, joining the locals is nearly as good.

Economic takeaway. Household local bias is not only a behavioral familiarity phenomenon. The evidence is consistent with local investors having useful information about nearby firms, especially smaller and less widely followed firms.

2 Incremental contribution of the paper

- **Moves from professionals to households.** Coval and Moskowitz show that mutual fund managers earn higher returns on local investments. This paper asks whether individual retail investors also appear to benefit from local information.
- **Separates familiarity from information.** The key distinction is whether local holdings earn superior returns. Familiarity alone predicts local concentration but not local outperformance.
- **Uses granular investor geography.** The data include household zip codes and firm headquarters, allowing the authors to measure household-stock distance directly rather than using coarse regional proxies.

- **Uses S&P 500 status as an information-asymmetry split.** Widely followed S&P 500 firms should have less scope for local informational advantage. Non-S&P 500 firms should have more scope.
- **Connects household choice to returns.** The paper does not only document local bias; it tests whether the local shares in household portfolios predict raw and risk-adjusted future returns.
- **Introduces relative local ownership.** Relative local ownership, or RLO, measures whether a stock is disproportionately held by nearby households after adjusting for the local distribution of investor wealth.
- **Provides mimicking strategies.** The paper asks whether an outside investor can exploit the information in household local choices by going long local positions, short nonlocal positions, or by favoring nonlocal stocks that local investors also hold heavily.
- **Extensive robustness.** The authors address geography, California/technology concentration, industry composition, retirement-account/own-employer concerns, risk adjustment, household locality, and alternative locality cutoffs.

3 Data and measurement

3.1 Household brokerage data

The primary data source is a large discount-broker dataset covering monthly positions and trades for about 78,000 U.S. households from 1991 to 1996. The data also contain household zip codes and some demographic information.

- **Asset focus.** The paper focuses on common stocks traded on the NYSE, AMEX, and Nasdaq.
- **Household location.** The authors use household zip codes and the Census Gazetteer to obtain latitude and longitude.
- **Firm location.** Firm headquarters come from COMPUSTAT county codes; stock returns and prices come from CRSP.
- **Sample restrictions.** The paper excludes households outside the continental United States, military zip codes, stocks not matched to CRSP and COMPUSTAT, and stocks not headquartered in the continental United States.
- **Baseline household sample.** At the end of 1991, the central sample contains 34,517 households with at least \$1,000 in common-stock holdings.
- **Portfolio composition.** The median household holds only two stocks, and the mean number of stocks is 3.3, so household portfolios are concentrated.

Interpretation. The sample is not a complete view of household wealth, because investors could hold assets at other brokers or in employer retirement plans. But it provides unusually detailed information on individual stock portfolios and household geography.

3.2 Distance and locality

The distance between a household and a firm is calculated using the latitude and longitude of the household zip code and firm headquarters. A stock is local to a household if the firm is headquartered within 250 miles of the household.

The paper also examines a 100 km cutoff as a robustness check. The authors describe the 250-mile cutoff as a plausible upper bound for the span of local information, while noting that it is arbitrary and perhaps conservative.

- **Distance to portfolio.** A value-weighted average distance from the household to the stocks in its portfolio.
- **Distance to market.** A value-weighted average distance from the household to the market portfolio.
- **Local portfolio share.** The fraction of the household's stock portfolio value invested in firms within 250 miles.
- **Market local share.** The fraction of all firms located within 250 miles of the household.
- **Relative locality.** The household's local portfolio share minus the local share of the market available near that household.

Interpretation. Relative locality matters because investors living near dense corporate clusters mechanically have more nearby firms. Subtracting the local market share helps separate local preference from local opportunity.

3.3 S&P 500 membership as an information-asymmetry proxy

The paper treats S&P 500 firms as more widely known and more closely followed. Local informational advantage should be smaller for these firms. Non-S&P 500 firms are less visible and likely have larger information asymmetries between local and nonlocal investors.

This split generates four portfolio categories:

$$\text{Local S\&P 500, } \quad \text{Local non-S\&P 500, } \quad \text{Nonlocal S\&P 500, } \quad \text{Nonlocal non-S\&P 500.} \quad (1)$$

Interpretation. The key prediction is not merely that local stocks outperform nonlocal stocks. The stronger prediction is that the local-minus-nonlocal return gap should be larger among non-S&P 500 stocks.

3.4 Relative local ownership

For firm j in year t , the paper defines relative local ownership as

$$RLO_{j,t} = \frac{\$own_{j,t}^{L(j)}}{\$own_{j,t}^{L(j)} + \$own_{j,t}^{NL(j)}} - \frac{\sum_k \$own_{k,t}^{L(k)}}{\sum_k \$own_{k,t}^{L(k)} + \sum_k \$own_{k,t}^{NL(k)}}. \quad (2)$$

The first term is the fraction of sample household ownership of stock j held by households located within 250 miles of that firm. The second term is the fraction of aggregate household portfolio wealth located within 250 miles of the firm. The paper restricts the RLO analysis to firms held by at least five investors in a given year.

Interpretation. RLO is high when local investors hold a stock more heavily than their local wealth share would predict. This makes RLO a possible crowd-based signal of local investor information.

3.5 Return measures

The paper uses both raw and risk-adjusted returns.

- **Raw household returns.** Household portfolio returns are calculated over one-year, three-year, and five-year horizons, using monthly rebalancing.
- **Four-factor alpha.** Household excess returns are regressed on the market, size, value, and momentum factors. The intercept is the average risk-adjusted monthly return.
- **Aggregate portfolios.** The authors aggregate household positions into local and nonlocal portfolios from February 1991 to December 1996.
- **Zero-cost portfolios.** The main mimicking portfolio is long local holdings and short nonlocal holdings, with analogous versions for S&P 500 and non-S&P 500 stocks.

Interpretation. The risk-adjusted specifications ask whether local outperformance survives standard return controls. The zero-cost portfolios ask whether the information in local holdings can be harvested without owning the original household portfolios.

4 Methodology and empirical specifications

4.1 Household locality determinants

The paper first asks which households invest more locally. It estimates cross-sectional regressions of household portfolio locality on household and portfolio characteristics:

$$\text{HHP locality}_i = \alpha + \beta_1 \log(\text{Number of stocks in HHP}_i) + \beta_2 \log(\text{Value of stocks in HHP}_i) + \beta_3 \log(\text{Household income}_i) \quad (3)$$

The regressions are estimated both with OLS and with zip-code fixed effects.

Interpretation. More diversified households should be less local if local bias partly reflects concentrated stock picking. Wealthier or higher-income households may be more local if they have more resources to collect information.

4.2 Firm-level relative local ownership regressions

The paper next asks which firms have higher RLO. It estimates pooled firm-year regressions of RLO on firm characteristics including leverage, market value, market-to-book, number of employees, S&P

500 membership, technology or biotechnology status, past returns, and risk.

- The sample includes seven pooled cross sections from early 1991 through November 1996.
- The regressions are estimated for all firms, S&P 500 firms, and non-S&P 500 firms.
- The paper reports both mean regressions and median regressions.

Interpretation. The RLO regressions show whether local ownership is concentrated in firms where local information should plausibly matter more.

4.3 Raw-return household regressions

The first return specification relates household returns to local and nonlocal portfolio shares:

$$R_{t,t+k}^{HHP} = \beta_L \% L_t + \beta_{NL} \% NL_t. \quad (4)$$

The expanded specification separates S&P 500 from non-S&P 500 holdings:

$$R_{t,t+k}^{HHP} = \beta_{L,SP} \% LSP_t + \beta_{NL,SP} \% NLSP_t + \beta_{L,NSP} \% LNSP_t + \beta_{NL,NSP} \% NLNSP_t. \quad (5)$$

The main comparisons are

$$D(SP) = \beta_{L,SP} - \beta_{NL,SP}, \quad D(NSP) = \beta_{L,NSP} - \beta_{NL,NSP}. \quad (6)$$

Interpretation. A positive $D(NSP)$ indicates that local non-S&P 500 holdings outperform nonlocal non-S&P 500 holdings. If $D(NSP) > D(SP)$, the local advantage is stronger where information asymmetry should be larger.

4.4 Four-factor risk adjustment

For each household with sufficient monthly returns, the paper estimates a Carhart four-factor regression:

$$R_t^{HHP} - R_{f,t} = \alpha^{HHP} + \beta_1 RMRF_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t^{HHP}. \quad (7)$$

It then regresses the household alpha on the same locality shares:

$$\alpha^{HHP} = \beta_L \% L + \beta_{NL} \% NL, \quad (8)$$

or on the four S&P/locality buckets.

Interpretation. If local non-S&P 500 stocks still outperform after factor adjustment, the result is less likely to be driven by local households simply holding small, value, or momentum stocks.

4.5 Aggregate local-minus-nonlocal portfolios

The paper constructs aggregate portfolios by combining household positions across the sample.

- Portfolio 1 is the value-weighted return on all household local holdings.
- Portfolio 2 is the value-weighted return on all household nonlocal holdings.
- Portfolio 3 is the zero-cost difference: long local and short nonlocal.
- The same construction is repeated for S&P 500 stocks and non-S&P 500 stocks.

The authors then split households by whether their own portfolios are below or above the median in relative locality.

Interpretation. This portfolio approach asks whether a strategy based only on geography and index membership would earn abnormal returns. The household-locality split asks whether the most locally oriented investors are the main source of the signal.

4.6 RLO and the value of following locals

The final major test asks whether nonlocal investors can use local ownership as a signal. The paper splits nonlocal non-S&P 500 stocks into high-RLO and low-RLO stocks.

- Local non-S&P 500 stocks are compared with nonlocal non-S&P 500 stocks with high RLO.
- Nonlocal non-S&P 500 high-RLO stocks are compared with nonlocal non-S&P 500 low-RLO stocks.

Interpretation. If high-RLO nonlocal stocks perform like local stocks, then observing the behavior of local investors can substitute for being local.

5 Identification and interpretation

5.1 What the paper can identify

The paper provides reduced-form evidence that household local portfolios contain value-relevant information. It does not observe the exact private signals investors use. The evidence is strongest when the return pattern lines up with the predicted information asymmetry pattern: large for non-S&P 500 stocks and weak or absent for S&P 500 stocks.

5.2 Familiarity versus information

Local bias by itself could be a familiarity or behavioral phenomenon. The performance tests are the key distinction. If households only buy nearby names because they recognize them, local holdings should not reliably outperform. The observed local outperformance, especially in non-S&P 500 stocks, supports an information-based interpretation.

5.3 Geographic confounds

A concern is that households in certain places, especially California during the technology-heavy 1990s, had local portfolios tilted toward industries that happened to perform well. The paper addresses this concern by using zip-code fixed effects, controlling for industry portfolio shares, excluding California households, analyzing California separately, and reporting region-level tests.

5.4 Risk and style confounds

Local stocks may be smaller or riskier. The four-factor regressions control for market, size, book-to-market, and momentum exposure. The key non-S&P 500 locality premium remains economically and statistically meaningful after these adjustments.

5.5 Own-employer and insider-information concerns

The data do not include 401(k) holdings, but households may hold employer or former-employer stock in other accounts. The paper checks this by excluding households with retirement accounts and by comparing households' largest local holdings to their other local holdings. The largest local holdings do not outperform the other local holdings, weakening the own-employer/inside-information explanation.

5.6 Measurement of locality

The 250-mile cutoff is plausible but arbitrary. The paper reports that using a 100 km cutoff makes the locality results even stronger, suggesting that 250 miles may be conservative.

5.7 Limits

- The brokerage data do not contain all household wealth.
- Headquarters location may not capture all operating locations or sources of local information.
- The analysis cannot observe the actual information households possess.
- The sample period is short and includes the early-to-mid 1990s, when regional and industry performance patterns were unusual in some places.
- The paper is careful about return predictability, but the strategy may be difficult to implement fully once trading costs, short-sale frictions, and real-time portfolio observability are considered.

6 Tables and figures

6.1 Table I and Figure 1: Household portfolios and the geography of firms/investors

Read Table I:

- The baseline sample contains 34,517 households with at least \$1,000 in stock holdings at the end of 1991.
- The median stock portfolio value is \$12,416, and the mean is \$32,911.
- The average household holds only 3.3 stocks, with a median of two stocks.
- The average distance from household to portfolio is 917 miles, while the average distance to the market portfolio is 1,225 miles.
- On average, 31.5% of the portfolio is within 250 miles, while only 12.6% of all firms are within 250 miles.
- The average local portfolio share exceeds the local market share by 18.9 percentage points.

Read Figure 1:

- The black points show publicly traded firm headquarters, and the gray points show retail investors.
- Firms and households cluster heavily in the Northeast, California, Florida, and other urbanized regions.
- Because both investors and firms are geographically clustered, the analysis needs to control for the local opportunity set.

Economic message: Households are not simply close to their portfolios because firms are distributed unevenly across the United States. Even after comparing household local shares to nearby market opportunities, there is a large local tilt.

Summary Statistics of Household Income and 1991 Household Portfolios			
The sample consists of 34,517 households that held at least \$1,000 of stocks in their portfolios at the end of 1991. Panel A presents the income statistics. Households could report annual income in nine ranges (0–15, 15–20, 20–30, 30–40, 40–50, 50–75, 75–100, 100–125, and 125+ thousands of dollars). One-eighth of the households did not provide income information. The income level for the households in the first eight categories is assumed to be the midpoint of the income range. Households that reported income greater than \$125,000 are assigned an income level of \$250,000 because calculations using the 1992 Survey of Consumer Finances suggest that the average household income in 1991 for households that had at least \$125,000 of income was \$250,000. Panel B presents the basic household portfolio statistics. Panel C presents the household portfolio statistics related to locality of households' investments. Household portfolio distance measures are computed by value-weighting the distance measures across individual stocks in the portfolio according to their household portfolio equity position.			
	Mean	Median	25–75th Percentile
Panel A: Income Statistics			
Income < \$50,000	25%		
Income between \$50,000 and \$100,000	39%		
Income over \$100,000	23%		
Household did not report income	12%		
Income level in thousands (conditional on reporting a range)	\$89	\$63	\$45–113
Panel B: Basic Portfolio Statistics			
Stock portfolio value	\$32,911	\$12,416	\$6,225–27,938
Number of stocks in portfolio	3.3	2	1–4
Household holds assets other than common stocks	38%		
Ratio of other assets to total assets in portfolio (conditional on holding other assets)	36%	30%	11–57%
Panel C: Distance Portfolio Statistics			
Distance to portfolio (miles)	917	796	311–1,391
Distance to market portfolio	1,225	1,030	768–1,810
Distance to portfolio–distance to market portfolio	–308	–251	–705–141
Percent of portfolio within 250 miles	31.5	2.8	0.0–65.6
Percent of all firms within 250 miles	12.6	5.7	4.3–15.1
Percent of portfolio within 250 miles–percent of all firms	18.9	–0.8	–5.7–48.6

Table I: Summary statistics of household income and 1991 portfolios.

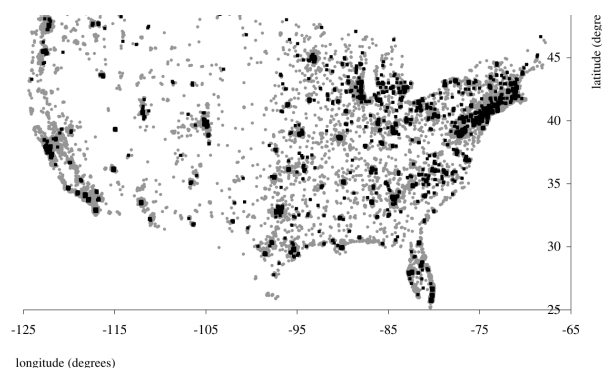


Figure 1. Distribution of U.S. publicly traded firms (black) and retail investors (gray),
Figure 1: Geographic distribution of public firms and retail investors.

6.2 Tables II and III: Locality in household portfolios

Read Table II:

- Actual household portfolios allocate 31.5% locally, compared with 12.6% under the market portfolio benchmark and 13.4% under the industry-portfolio benchmark.
- Local non-S&P 500 stocks are especially overweighted: 13.9% of actual household portfolios versus 3.1% under the market portfolio.
- Households underweight nonlocal S&P 500 stocks relative to the market portfolio.
- California households are somewhat more local than non-California households, but the differences are modest relative to the overall local bias.
- 17.0% of households invest only locally, while 48.6% invest only nonlocally.

Read Table III:

- More diversified households invest less locally: the number of stocks and the indicator for holding other assets are negatively related to locality.
- Larger stock portfolios are more local, which may reflect greater resources or willingness to collect information.
- Household income is positively related to locality in OLS, though the fixed-effect evidence is weaker.
- Zip-code fixed effects help control for the spatial distribution of firms and households.

Economic message: Locality is not random. It is stronger for concentrated portfolios and for households with larger stock portfolios, consistent with local investing being a deliberate stock-picking choice rather than a diversified passive allocation.

Table II
Composition of Household Portfolios by Locality, 1991

The table reports the average local and nonlocal portfolio shares across 34,517 households with at least \$1,000 in stock holdings at the end of 1991. In computing the averages, portfolio shares are weighted by the total household portfolio position in common stocks. Portfolio shares are rounded to one-tenth of a percent. Thus, shares may not add up to 100% due to rounding errors. A stock is defined as local for a given household if the firm is headquartered within 250 miles of the household. Panel A presents the basic breakdown between local and nonlocal stocks, whereas panel B presents a more detailed breakdown that simultaneously considers the stock's locality and the S&P 500 status. The first column reports the hypothetical shares of household portfolios under the assumption that all households invested in the market portfolio. The second column presents the hypothetical locality breakdown wherein for each stock the measure of locality is the ratio of aggregate market capitalization of all local firms belonging to the same industry and aggregate market capitalization of all firms nationwide belonging to the same industry, with all the individual stock measures then aggregated according to the weights of the respective stocks in the household portfolio. Thus, the portfolio shares listed for this benchmark represent the hypothetical locality breakdown if, instead of holding its individual stocks, each household invested in the appropriate value-weighted industry portfolio corresponding to each of its individual stocks. The third column presents the average portfolio shares for all households in the sample. Tests of statistical significance reported along the first two columns pertain to the test of equality of portfolio shares with portfolio shares in the third column, respectively. The fourth column presents the average portfolio shares of all households except for California households, whereas the fifth column presents the average portfolio shares of California households. Tests of statistical significance reported along the fourth column pertain to the test of equality of portfolio shares with portfolio shares in the fifth column. The sixth and seventh columns present the average portfolio shares of households with portfolio values of at least \$10,000 and at least \$25,000 (corresponding roughly to the median and the top quartile household portfolio sizes), respectively. The last column presents the fraction of the households in the sample that are invested only in the category highlighted in the respective row of the table.

	If Invested in Market Portfolio	If Invested in Industry Portfolio	All Households	Exclude CA Households	CA Households	Portfolio ≥ \$10,000	Portfolio ≥ \$25,000	Fraction Invested Only in Category
Panel A: Basic Breakdown								
Local	12.6%*	13.4%*	31.5%	30.8%*	34.0%	31.0%	29.2%	17.0%
Nonlocal	87.4%*	86.6%*	68.5%	69.2%*	66.0%	69.0%	70.7%	48.6%
All stocks	100%	100%	100%	100%	100%	100%	100%	
Panel B: Breakdown with Respect to S&P 500								
Local (S&P 500)	9.5%*	10.0%*	17.6%	17.5%	17.7%	18.6%	18.1%	8.3%
Local (not S&P 500)	3.1%*	3.9%*	13.9%	13.1%*	16.3%	12.5%	11.2%	6.8%
Nonlocal (S&P 500)	60.1%*	62.1%	42.2%	41.8%	45.1%	47.0%	47.0%	20.2%
Nonlocal (not S&P 500)	24.9%*	26.3%	26.4%	27.0%*	24.2%	23.9%	23.7%	13.3%
All stocks	100%	100%	100%	100%	100%	100%	100%	
Mean number of stocks			3.3	3.4**	3.1	4.3	5.8	
Number of households	34,517	34,517	34,517	25,151	9,366	20,045	9,581	34,517

*Significance of tests of equality of coefficients from the first two columns with the respective coefficient in the third column at the 1% level.
**Significance of tests of equality of coefficients from the Exclude CA Households column with the respective coefficient in the CA households column at the 1% level.

Table II: Composition of household portfolios by locality and S&P 500 status.

6.3 Tables IV and V: Relative local ownership and the first return tests

Read Table IV:

Table III
Regression of Locality of Household's Portfolio upon Household and Portfolio Characteristics, 1991

The table presents the results of fitting cross-sectional regressions of four household portfolio (HHP) locality measures, computed at the end of 1991, on several explanatory variables according to equation (2) from Section II.B. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. The first two measures are based on the average distance from the household to its portfolio. The other two measures are based on the fraction of portfolio value within 250 miles from the household. Each specification is fitted in both the OLS and the fixed-effects regression frameworks and the corresponding results are presented in the two columns associated with the locality measure. The intent of the fixed-effect regression, carried out at the zip-code level, is to control for potential biases in the geographic distribution of households and firms in the sample. The first two measures are based on the average distance from the household to its portfolio and thus are negatively related to household portfolio locality. By contrast, the other two measures are based on the fraction of portfolio value within 250 miles from the household and are each positively related to household portfolio locality. For those households that did not report income, log(Household income) is set to zero and the indicator variable Did not report income is set to one. Standard errors (shown in parentheses) allow for heteroskedasticity.

	Average Distance to Portfolio Stocks (Miles)	Average Distance to Portfolio Stocks (Miles) - Distance to Market	Percentage of Portfolio Value within 250 Miles (Local)	Percentage of Portfolio Local - Percentage of Market Local
log(Number of stocks in HHP)	100.0*** (5.5)	117.9*** (6.8)	131.4*** (5.4)	-6.7*** (0.3)
log(Value of stocks in HHP)	-16.1*** (3.6)	-22.4*** (4.5)	-32.5*** (3.6)	1.5*** (0.2)
log(Household income)	-10.6*** (5.1)	-6.1 (7.0)	2.6*** (5.1)	0.8* (0.4)
Did not report income	0.3 (24.3)	16.6 (32.9)	7.1*** (23.9)	0.3 (1.4)
Hold other assets	65.9*** (7.6)	60.8*** (9.5)	71.2*** (9.5)	-4.0*** (0.4)
Constant	1.018*** (.39)	-42.8 (28.3)	14.1*** (2.3)	-3.8*** (2.2)
Zip-code effects	No	Yes	No	Yes
R ²	0.0167	0.0210	0.0255	0.0208
Number of households	34,517	34,517	34,517	34,517

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table III: Household portfolio locality and household/portfolio characteristics.

- Relative local ownership is negatively related to leverage and positively related to the number of employees.
- S&P 500 membership is strongly negatively related to RLO in the all-firm regressions.
- Technology or biotechnology status is positively related to RLO in median regressions.
- The firm-level determinants are stronger for non-S&P 500 firms than for S&P 500 firms.

Read Table V:

- In Panel A, local holdings outperform nonlocal holdings by 3.2% over the next year and by 4.2% with zip-code fixed effects.
- In Panel B, the S&P 500 local-minus-nonlocal difference is small and often insignificant.
- The non-S&P 500 local-minus-nonlocal difference is much larger: 6.2% over the next year, or 6.7% with zip-code fixed effects.
- In the four-factor alpha columns, local non-S&P 500 holdings earn 25.5 basis points per month, while nonlocal non-S&P 500 holdings earn 4.6 basis points.
- The risk-adjusted non-S&P 500 locality difference is 20.9 basis points per month, and the difference-in-differences relative to S&P 500 stocks is 23.9 basis points.

Economic message: The first return tests are the core evidence that local bias contains information. The premium appears exactly where the information story predicts: local non-S&P 500 stocks.

Relation between Relative Local Ownership (RLO) and Firm Characteristics

The table presents the results of fitting cross-sectional regressions of the relative local ownership (*RLO*) measure on several explanatory variables (see Section III for details). *RLO* is defined as the fraction of stock ownership in the firm located within 250 miles of the firm minus the fraction of total nationwide household portfolio value located within 250 miles of the firm (equation (3)). Market value and the Number of employees are each expressed in logarithms. The sample is restricted to stocks owned by at least five households in the sample at the beginning of each year (end of January for 1991) and at the end of November 1996, and consists of seven pooled cross sections from 1991 to November 1996. Regressions are carried out for all firms, S&P 500 firms, and non-S&P 500 firms. For each of the three groups of stocks the table reports results of OLS regressions and median regressions (columns labeled as Mean and Median, respectively). Standard errors (shown in parentheses) allow for heteroskedasticity as well as correlation across observations of the same firm over time. Standard errors for the median regressions are based on 200 bootstrap replications.

	All Firms		S&P 500 Firms		Non-S&P 500 Firms	
	Mean	Median	Mean	Median	Mean	Median
Leverage (debt to assets)	-6.9*** (2.0)	-5.5*** (1.2)	-2.1 (7.4)	-2.0 (3.9)	-7.8*** (2.0)	-7.8*** (1.5)
Market value	-0.3 (0.3)	0.6*** (0.2)	-1.0 (0.9)	1.1** (0.5)	-0.2 (0.3)	0.9*** (0.3)
Market-to-book ratio	-0.0 (0.1)	-0.0 (0.1)	-0.6 (0.8)	-1.2*** (0.4)	-0.0 (0.1)	-0.0 (0.1)
Number of employees	1.1*** (0.3)	1.1*** (0.2)	-1.0 (0.8)	-1.9*** (0.5)	1.2*** (0.3)	1.5*** (0.2)
S&P 500	-5.4*** (1.2)	-5.4*** (0.6)				
Tech. or biotech.	0.3 (0.8)	2.1*** (0.6)	0.5 (1.6)	2.0** (0.9)	0.4 (0.9)	2.4*** (0.6)
Return past 1 year	0.2 (0.3)	-0.1 (0.3)	1.5 (1.0)	-0.1 (1.3)	0.1 (0.3)	-0.2 (0.3)
Total risk past 1 year	-4.8 (3.4)	-3.2 (3.5)	2.5 (19.1)	26.3** (12.3)	-3.5 (3.2)	-0.5 (3.4)
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0088	0.0073	0.0155	0.0075	0.0093	0.0115
Number of observations	16,057	16,057	3,021	3,021	13,036	13,036

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table IV: Relative local ownership and firm characteristics.

Table V
Relation between Household Portfolio Returns and Locality

The table presents the results of fitting cross-sectional regressions of household portfolio returns on portfolio locality shares (computed at the end of 1991). The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. Household raw returns (calculated based on monthly rebalancing of their portfolio) are calculated over the next year, next 3 years, and next 5 years and are expressed in percentages. These returns are then related to the portfolio shares, computed at the end of 1991, according to the specifications outlined in equations (4) and (5). The results are presented in the first six columns. For each investment horizon, the table reports both the OLS regression and the fixed-effects regression at the zip-code level. The last two columns report results of fitting cross-sectional regressions of average excess monthly household returns α (calculated from a four-factor model according to equation (6) and expressed in basis points) on the portfolio shares computed at the end of 1991 according to the specifications outlined in equation (7). Standard errors (shown in parentheses) allow for heteroskedasticity.

	1 Year Raw Returns (%)	3 Year Raw Returns (%)	5 Year Raw Returns (%)	Average Monthly Excess Returns (bp)				
Panel A: Breakdown by Locality (250 Miles)								
Share of local stocks (L)	13.7*** (0.5)	14.5*** (0.7)	26.4*** (1.4)	29.3*** (2.1)	110.7*** (3.9)	114.3*** (7.5)	17.1*** (1.8)	16.1*** (2.3)
Share of nonlocal stocks (NL)	10.6*** (0.3)	10.3*** (0.4)	22.3*** (0.8)	20.9*** (1.0)	97.8*** (2.5)	96.1*** (3.6)	10.3*** (1.0)	10.7*** (1.2)
Locality difference = L - NL	3.2*** (0.7)	4.2*** (0.9)	4.1*** (1.7)	9.4*** (2.8)	12.9*** (4.9)	18.2*** (10.5)	6.8*** (2.2)	5.4*** (3.1)
Zipcode effects	No	Yes	No	Yes**	No	Yes***	No	Yes
R ²	0.0011	0.0017	0.0006	0.0023	0.0012	0.0023	0.0004	0.0002
Number of observations	27,032	27,032	12,067	12,067	6,462	6,462	26,093	26,093

(continued)

Table V, part 1: Local versus nonlocal raw returns.

6.4 Tables V continued and VI: Risk-adjusted returns and robustness checks

Read Table V continued:

- The breakdown by S&P 500 membership shows that the return to locality is not uniform.
- Local S&P 500 stocks do not reliably outperform nonlocal S&P 500 stocks after risk adjustment.
- Local non-S&P 500 stocks outperform nonlocal non-S&P 500 stocks across raw and risk-adjusted measures.

Read Table VI:

- Adding 14 industry portfolio shares reduces the non-S&P 500 alpha difference only modestly, from 20.9 to 17.3 basis points per month.
- Excluding California households still produces a positive non-S&P 500 locality premium.
- California households have a large raw locality premium, but after industry controls the magnitude becomes similar to non-California households.
- Excluding households with retirement accounts does not weaken the results; the non-S&P 500 locality premium remains large.

Economic message: The evidence is not just a California or technology-sector artifact. The local non-S&P 500 premium survives industry controls, California exclusions, and retirement-account exclusions.

Table V—Continued

	1 Year Raw Returns (%)		3 Year Raw Returns (%)		5 Year Raw Returns (%)		Average Monthly Excess Returns (bp)	
Panel B: Breakdown by Locality (250 Miles) and S&P 500 Membership								
Share of S&P 500 local stocks (SP, L)	10.8***	11.8***	21.9***	24.5***	100.8***	103.0***	10.8***	9.0***
Share of S&P 500 nonlocal stocks (SP, NL)	10.5***	9.8***	(0.1)	(2.6)	(3.9)	(9.3)	(1.9)	(2.7)
Share of S&P 500 nonlocal stocks (NSP, NL)	10.1***	9.7***	21.7***	20.3***	99.3***	99.8***	13.8***	14.5***
S&P 500 locality diff. (DSP) = SP, L – SP, NL	0.3	2.0**	0.2	4.1	1.5	3.1	–3.0	–5.5
Share of non-S&P 500 local stocks (NSP, L)	0.8	2.1**	0.2	4.1	9.5**	3.1	–3.0	–5.5
D(NSP) = NSP, L – SP, NL	(0.6)	(1.0)	(1.7)	(3.3)	(4.8)	(11.7)	(2.3)	(3.4)
Share of non-S&P 500 nonlocal stocks (NSP, NL)	17.8***	17.9***	31.5***	36.6***	128.0***	131.2***	25.5***	23.9***
Share of non-S&P 500 local stocks (NSP, L)	(1.1)	(1.4)	(2.9)	(4.3)	(8.4)	(15.9)	(3.4)	(4.2)
Share of non-S&P 500 nonlocal stocks (NSP, NL)	11.5***	11.2***	23.4***	22.1***	94.4***	88.4***	4.6**	5.1*
Non-S&P 500 locality diff. (D(NSP) = NSP, L – NSP, NL)	0.0	6.7***	8.1***	14.5***	33.6***	44.8***	20.9***	19.9***
D(NSP) = NSP, L – NSP, NL	1.6	(1.4)	(1.9)	(3.6)	(5.4)	(11.2)	(19.9)	(4.3)
Difference between the two differences = D(NSP) – DSP	5.0***	4.6**	7.9***	10.4	32.1***	41.7*	23.9***	25.4***
Difference between the two differences = D(NSP) – DSP	(1.6)	(2.1)	(4.1)	(6.5)	(12.2)	(22.9)	(5.0)	(6.8)
Zip-code effects	No	Yes	No	Yes**	No	Yes**	No	Yes
R ²	0.0051	0.0053	0.0023	0.0042	0.0050	0.0051	0.0017	0.0016
Number of observations	27,032	27,032	12,067	12,067	6,462	6,462	26,093	26,093

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table V, part 2: Locality, S&P 500 status, and risk-adjusted returns.

6.5 Tables VII and VIII: Regional robustness and household-locality portfolios

Read Table VII:

- The one-year local-minus-nonlocal raw-return difference is positive in eight of nine broad Census regions.
- Local non-S&P 500 stocks outperform nonlocal non-S&P 500 stocks in all nine broad regions.

Table VI
Robustness Checks on the Relation between Household Portfolio Monthly Excess Returns and Locality

The table presents the robustness checks on the relation between household portfolio monthly excess returns and locality (reported in the last two columns of Table V). The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. The table presents key results of the fitting cross-sectional regressions of average excess monthly household returns r (calculated from a four-factor model according to equation (6) and expressed in basis points) on the portfolio shares, computed at the end of 1991, according to the specifications outlined in equation (7). Regressions are fitted for the full sample of households, all households except for California households, California households, and for households that did not have retirement accounts, respectively. For each group of households, the table also documents the regressions that feature portfolio shares across 14 industry groups at the end of 1991. The table reports the difference in estimated coefficients. For the basic locality regression, the top panel, $\beta_{L,SP} - \beta_{NL,SP}$, is reported. For the regressions that interact locality with S&P 500 membership, displayed in the bottom panel, $\beta_{L,SP} - \beta_{NL,SP}$, $\beta_{L,NSP} - \beta_{NL,NSP}$, and $\beta_{L,NSP} - \beta_{NL,NSP} - \beta_{NL,NSP} - \beta_{NL,NSP}$ are reported. Standard errors (shown in parentheses) allow for heteroskedasticity.

	Full Sample		Exclude CA Households		CA Households		Exclude Households With Retirement Accounts	
Panel A: Breakdown by Locality (250 Miles)								
Locality difference = L – NL	6.8***	4.9*	9.4***	8.1***	0.7	-7.4	8.8***	4.3
	(2.2)	(2.2)	(2.5)	(2.5)	(4.4)	(4.6)	(3.1)	(3.1)
Portfolio industry shares included	No	Yes	No	Yes	No	Yes	No	Yes
R ²	0.0004	0.0217	0.0008	0.0250	0.0000	0.0245	0.0007	0.0267
Number of observations	26,093	26,093	19,063	19,063	7,030	7,030	13,128	13,128
Panel B: Breakdown by Locality (250 Miles) and S&P 500 Membership								
S&P 500 locality diff. D(SP) = SP, L – SP, NL	-3.0	-5.1**	4.6*	2.5	-22.6***	-25.9***	-2.4	-6.4**
	(2.3)	(2.3)	(2.7)	(2.6)	(4.5)	(5.0)	(3.2)	(3.2)
Non-S&P 500 locality diff. D(NSP) = NSP, L – NSP, NL	20.9***	17.5***	16.5***	16.7***	32.1***	16.8**	24.5***	20.0***
	(4.3)	(4.3)	(5.0)	(5.0)	(8.4)	(8.1)	(6.1)	(6.1)
Difference between the two differences = D(NSP) – D(SP)	23.9***	22.4***	11.9**	14.2***	54.7***	42.7***	27.2***	26.4***
	(4.8)	(4.9)	(5.7)	(5.7)	(9.6)	(9.9)	(7.0)	(7.0)
Portfolio industry shares included	No	Yes	No	Yes	No	Yes	No	Yes
R ²	0.0017	0.0232	0.0012	0.0260	0.0063	0.0286	0.0023	0.0285
Number of observations	26,093	26,093	19,063	19,063	7,030	7,030	13,128	13,128

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table VI: Robustness checks on monthly excess returns and locality.

- The result is not only a California result. The table separately reports Pacific excluding California and California.
- Some regional estimates are noisy, but the overall pattern is broad.

Read Table VIII:

- Households are sorted into four portfolios by relative locality.
- The most local quartile has the highest average raw monthly return, 1.582%.
- The lower three quartiles have similar performance; the top-locality quartile stands out.
- In risk-adjusted returns, the most local quartile has an average excess monthly return of 0.154%.

Economic message: The regional tests show that the result is not confined to one area. The portfolio-locality sort suggests that the benefit of local information is concentrated among households that commit most strongly to local investing.

Table VII
Relation between Household Portfolio Returns and Locality by Region

The table presents the robustness checks on the relation between household portfolio returns and locality (reported in Table V) by region. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. Household raw returns are calculated over the next year and are expressed in percentages. This return is then related to the portfolio shares, computed at the end of 1991, according to the specifications outlined in equations (4) and (5). Aside from regressions involving raw returns, the table presents the results of fitting cross-sectional regressions of average excess monthly household returns α (calculated from a four-factor model according to equation (6) and expressed in basis points) on the portfolio shares, computed at the end of 1991, according to the specifications outlined in equation (7). Households are aggregated by region, as defined by the U.S. Bureau of the Census. Alongside regression results, #HH denotes the number of households in each region. The table reports the differences in estimated coefficients for 1-year raw returns and average monthly excess returns across U.S. regions. For the basic locality regression, reported in the first and fourth columns, the difference $\beta_L - \beta_{NL}$ is reported. For the regressions that interact locality with S&P 500 membership, reported in the second and fifth columns, the difference $\beta_{L,NSP} - \beta_{NL,NSP}$ is reported. Standard errors (not shown) allow for heteroskedasticity.

Region	1 Year Raw Returns (%)			Average Monthly Excess Returns (bp)		
	$\beta_L - \beta_{NL}$	$\beta_{L,NSP} - \beta_{NL,NSP}$	#HH	$\beta_L - \beta_{NL}$	$\beta_{L,NSP} - \beta_{NL,NSP}$	#HH
New England: CT, ME, MA, NH, RI, VT	6.9**	22.0***	1,244	22.3**	15.5	1,232
Middle Atlantic: DE, DC, MD, NJ, NY, PA	2.1	8.1***	4,617	11.9*	10.9	4,456
South Atlantic: FL, GA, NC, SC, VA, WV	9.8***	8.6*	2,949	7.8	-1.6	2,804
East north central: IL, IN, OH, MI, WI	0.2	2.3	3,078	-21.0***	-24.6*	2,911
East south central: AL, KY, MS, TN	7.6**	17.1***	661	-1.0	52.7**	672
West north central: IA, KS, MN, MO, NE, ND, SD	-2.4	0.2	1,342	15.9	7.4	1,285
West south central: AR, LA, OK, TX	10.9***	14.9***	2,488	7.6	7.5	2,411
Mountain: AZ, CO, ID, MT, NV, NM, UT, WY	7.4*	4.6	1,738	10.4	26.7	1,743
Pacific: CA, OR, WA	1.9	4.1	8,915	10.4***	43.3***	8,579
Pacific—exclude CA	-7.8***	-9.0	1,634	49.3***	77.7***	1,549
CA	4.5***	8.4***	7,281	0.7	32.1**	7,030
All regions	3.2***	6.2***	27,032	6.8***	20.9***	26,093

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table VII: Locality-return relation by region.

The table documents the performance of household portfolios aggregated into four portfolios according to their locality. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. At the beginning of each year (end of January 1991), households are ranked based on the locality of their portfolio relative to the market portfolio (fraction of household portfolio within 250 miles minus the fraction of market portfolio within 250 miles) and then assigned to locality quartiles. The sample period is from February 1991 to December 1996. The average household relative locality cutoffs across all years are -6.3% for the first quartile, -0.5% for the second quartile, and 46.6% for the third quartile. The return for each quartile represents the weighted return for all the households in that quartile (the weight is the value of each household's portfolio). Raw returns are presented in panel A, whereas excess returns are presented in panel B. Excess monthly returns are calculated from a four-factor model, which accounts for the market, size, book-to-market, and momentum effects (Carhart (1997)). These four locality portfolios are re-balanced at the beginning of each year based on households' locality relative to the market portfolio. In each panel, aside from reporting the performance of the four locality portfolios, for the bottom three locality-quartile portfolios, the table also reports their performance differential relative to the most local quartile portfolio. Standard errors reported are Newey-West standard errors with autocorrelation up to three lags.

Household by Locality Quartile	Panel A: Raw Returns (%)		Panel B: Excess Returns (%)	
	Average Monthly Return	Difference Relative to Top Quartile	Average Excess Monthly Return	Difference Relative to Top Quartile
Quartile 1 (least local)	1.393*** (0.375)	-0.189 (0.120)	-0.003 (0.108)	-0.157 (0.103)
Quartile 2	1.344*** (0.333)	-0.238** (0.122)	0.016 (0.084)	-0.138 (0.103)
Quartile 3	1.361*** (0.331)	-0.221* (0.125)	0.015 (0.080)	-0.140 (0.103)
Quartile 4 (most local)	1.582*** (0.425)		0.154 (0.161)	

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

D. Pursuing Locality: Dissecting Portfolios by Locality

Table VIII: Performance of household portfolios sorted by locality.

6.6 Tables IX and X: Aggregate zero-cost portfolios and household locality

Read Table IX:

- The all-stock zero-cost portfolio, long local and short nonlocal, earns 0.17% per month in raw returns and 0.09% per month in alpha.
- The S&P 500 local-minus-nonlocal portfolio has small and statistically weak performance.
- The non-S&P 500 local-minus-nonlocal portfolio earns 0.25% per month in raw returns and 0.20% per month in alpha.

- The only portfolio formation criteria are geography and S&P 500 status.

Read Table X:

- The zero-cost strategy is essentially absent among households with below-median locality.
- Above-median locality households generate much stronger local-minus-nonlocal returns.
- For non-S&P 500 stocks among above-median locality households, the zero-cost portfolio earns 0.39% per month in raw returns and 0.28% per month in alpha.

Economic message: A simple local-minus-nonlocal strategy earns abnormal returns, and the effect is strongest among non-S&P 500 stocks held by households that themselves invest more locally. The signal is not just geography; it is geography combined with investor behavior.

Table IX
Regression of Monthly Portfolio Returns from February 1991 to December 1996, Local versus Nonlocal Investments

The table documents the performance of aggregate household portfolios constructed by aggregating all the households' local holdings and nonlocal holdings. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. The sample period is from February 1991 to December 1996. The average raw return per month and the excess return per month (Alpha) from various portfolios are expressed in percentage points. Excess monthly returns are calculated from a four-factor model, which accounts for the market, size, book-to-market, and momentum effects (Carhart (1997)). The loadings on each of the four factors are presented for nine portfolios. Portfolio 1 (All Stocks, Local) is the return on local investments weighted across all household portfolios. Portfolio 2 (All Stocks, Nonlocal) is the return on nonlocal investments weighted across all household portfolios. Portfolio 3 (All Stocks, Difference) is the return on the zero-cost portfolio that is long local stocks (Portfolio 1) and is short nonlocal stocks (Portfolio 2), again weighted across all household portfolios. The middle three portfolios are defined analogously, but just focus on S&P 500 stocks held by households. The right three portfolios are defined analogously, but just focus on non-S&P 500 stocks held by households. Standard errors reported are Newey-West standard errors with autocorrelation up to three lags.

	All Stocks			S&P 500 Stocks			Non-S&P 500 Stocks		
	Local	Nonlocal	Difference	Local	Nonlocal	Difference	Local	Nonlocal	Difference
Market	1.14*** (0.06)	1.07*** (0.03)	0.07* (0.04)	1.06*** (0.03)	1.03*** (0.03)	0.03 (0.08)	1.28*** (0.08)	1.15*** (0.06)	0.11* (0.06)
Small minus big	0.24*** (0.09)	0.15*** (0.05)	0.09*** (0.04)	-0.14*** (0.07)	-0.17*** (0.05)	0.03 (0.03)	0.84*** (0.11)	0.84*** (0.08)	-0.00 (0.06)
High minus low	-0.09 (0.07)	-0.10** (0.05)	0.01 (0.03)	-0.08 (0.08)	-0.09* (0.05)	0.01 (0.04)	-0.11 (0.07)	-0.12*** (0.05)	0.01 (0.04)
Up minus down	-0.08 (0.06)	-0.07** (0.04)	-0.01 (0.05)	-0.11* (0.06)	-0.12*** (0.04)	0.01 (0.03)	-0.05 (0.09)	0.03 (0.05)	-0.07 (0.05)
p-value of regression	0.000*** 0.10	0.000*** 0.02	0.000*** 0.09	0.000*** 0.22*	0.000*** 0.16*	0.111 0.06	0.000*** -0.06	0.000*** -0.26	0.305 0.20**
Raw return	0.13 (0.39)	0.09 (0.34)	0.07 (0.08)	0.15 (0.35)	0.09 (0.31)	0.11 (0.07)	0.18 (0.53)	0.17 (0.49)	0.10 (0.11)

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table IX: Aggregate local-minus-nonlocal portfolio returns.

Nonlocal) from February 1991 to December 1996, Formed with Respect to Household Portfolio Locality

The table documents the performance of aggregate household portfolios constructed by aggregating all the households' local holdings and nonlocal holdings while controlling for household locality. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. The sample period is from February 1991 to December 1996. The average raw return per month and the excess return per month (Alpha) from various portfolios are expressed in percentage points. Excess monthly returns are calculated from a four-factor model, which accounts for the market, size, book-to-market, and momentum effects (Carhart (1997)). The left panel (All Stocks) focuses on all stocks held by households, and breaks those holdings into local and nonlocal investments. The return on the zero-cost portfolio that is long local stocks and is short nonlocal stocks weighted across all household portfolios is displayed in the top row. This calculation is then repeated restricting the sample to households whose portfolio locality share is below the median (middle row) and above the median (bottom row). At the beginning of each year (end of January 1991), households are ranked based on the locality of their portfolio relative to the market portfolio (fraction of household portfolio within 250 miles minus fraction of market portfolio within 250 miles). The average household relative locality median cutoff across all years is -0.5% for the median. The three rows of the middle panel (S&P 500 Stocks) are defined analogously, but just focus on S&P 500 stocks held by households. The three rows of the right panel (Non-S&P 500 Stocks) are defined analogously, but just focus on non-S&P 500 stocks held by households. Standard errors reported are Newey-West standard errors with autocorrelation up to three lags.

	All Stocks		S&P 500 Stocks		Non-S&P 500 Stocks	
	Raw Return	Alpha	Raw Return	Alpha	Raw Return	Alpha
All households	0.17** (0.08)	0.09 (0.07)	0.11 (0.07)	0.06 (0.07)	0.25** (0.11)	0.20** (0.10)
Households by locality						
Household locality	0.00 (0.09)	-0.06 (0.09)	0.06 (0.10)	0.09 (0.11)	0.01 (0.18)	-0.06 (0.18)
below median	0.22** (0.09)	0.12 (0.08)	0.12 (0.08)	0.05 (0.08)	0.39*** (0.11)	0.28*** (0.10)
above median						

*** and ** indicate significance at the 1% and 5% levels, respectively.

Table X: Zero-cost portfolio returns by household-locality group.

6.7 Tables XI and XII: Employer-stock and California robustness

Read Table XI:

- The table compares a household's biggest local stock with its other local stocks.
- If the local premium were driven by own-employer or former-employer stock, the biggest local holding should outperform.
- The biggest local holdings do not outperform other local holdings in raw returns.
- In risk-adjusted returns, the biggest local holdings slightly underperform other local holdings.

Read Table XII:

- The table repeats the aggregate portfolio approach separately for non-California and California households.
- Non-California households still exhibit a positive non-S&P 500 local-minus-nonlocal return.
- California households also show strong local-minus-nonlocal performance, but the finding is not necessary for the full-sample result.

Economic message: The main results are hard to explain as employee stock or as a pure California/technology phenomenon. The local-information interpretation survives these alternative explanations.

Table XI
Regression of Monthly Local Stock Portfolio Returns from February 1991 to December 1996, Biggest Local versus Other Local Investments

The table documents the performance of aggregate local household portfolios. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local if the firm is headquartered within 250 miles of the household. The sample period is from February 1991 to December 1996. The average raw return per month and the excess return per month (Alpha) from various portfolios are expressed in percentage points. Excess monthly returns are calculated from a four-factor model, which accounts for the market, size, book-to-market, and momentum effects (Carhart (1997)). The loadings on each of the four factors are presented for nine portfolios. Portfolio 1 (All Local Stocks, Biggest Local Stock) is the return on the biggest local investment weighted across all household portfolios. Portfolio 2 (All Local Stocks, Other Local Stocks) is the return on local investments, other than the household's largest, weighted across all household portfolios. Portfolio 3 (All Local Stocks, Difference) is the return on the zero-cost portfolio that is long the biggest local stock (Portfolio 1) and is short the other local stocks (Portfolio 2), again weighted across all household portfolios. The middle three portfolios are defined analogously, but just focus on S&P 500 stocks held by households. The right three portfolios are defined analogously, but just focus on non-S&P 500 stocks held by households. Biggest S&P 500 Local Stock represents the household's biggest local investment if it happens to be a member of the S&P 500, and Biggest Non-S&P 500 Local Stock represents the household's biggest local investment if it happens to not be a member of the S&P 500. Standard errors reported are Newey-West standard errors with autocorrelation up to three lags.

	All Local Stocks			S&P 500 Local Stocks			Non-S&P 500 Local Stocks		
	Biggest	Other	Difference	Biggest	Other	Difference	Biggest	Other	Difference
Market	1.14*** (0.06)	1.15*** (0.05)	-0.01 (0.02)	1.05*** (0.06)	1.10*** (0.05)	-0.05*** (0.02)	1.28*** (0.09)	1.22*** (0.07)	0.06 (0.04)
Small minus big	0.25*** (0.09)	0.23*** (0.09)	0.01 (0.02)	-0.13** (0.07)	-0.17*** (0.08)	0.03 (0.03)	0.83*** (0.12)	0.87*** (0.12)	-0.04 (0.04)
High minus low	-0.11 (0.07)	-0.04 (0.07)	-0.07*** (0.02)	-0.09 (0.02)	-0.05 (0.02)	-0.04 (0.08)	-0.15* (0.08)	-0.02 (0.07)	-0.14*** (0.04)
Up minus down	-0.06 (0.06)	-0.14*** (0.06)	0.08*** (0.02)	-0.09 (0.06)	-0.18*** (0.02)	0.09*** (0.06)	-0.03 (0.09)	-0.10 (0.08)	0.07* (0.04)
p-value of regression	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***
Alpha	0.07 (0.14)	0.19* (0.11)	-0.11* (0.07)	0.19 (0.11)	0.30*** (0.11)	-0.11 (0.07)	-0.08 (0.20)	0.02 (0.17)	-0.10 (0.10)
Raw return	1.51*** (0.40)	1.50*** (0.38)	-0.08 (0.07)	1.46*** (0.35)	1.55*** (0.34)	-0.09 (0.07)	1.60*** (0.54)	1.65*** (0.51)	-0.05 (0.10)

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table XI: Biggest local holding versus other local holdings.

Table XII
Regression of Monthly Portfolio Returns from February 1991 to December 1996, Robustness to California

The table presents a robustness check of the performance of aggregate household portfolios (presented in Table IX) with respect to the California households. The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. The sample period is from February 1991 to December 1996. The average raw return per month and the excess return per month (Alpha) from various portfolios are expressed in percentage points. Excess monthly returns are calculated from a four-factor model, which accounts for the market, size, book-to-market, and momentum effects (Carhart (1997)). The loadings on each of the four factors are presented for six portfolios. Portfolio 1 (Non-S&P 500 Stocks, Local) is the return on local, non-S&P 500 local investments weighted across all non-California household portfolios. Portfolio 2 (Non-S&P 500, Non-local) is the return on nonlocal, non-S&P 500 investments weighted across all non-California household portfolios. Portfolio 3 (Non-S&P 500 Stocks, Difference) is the return on the zero-cost portfolio that is long non-S&P 500 local stocks (portfolio 1) and is short nonlocal stocks (portfolio 2), again weighted across all non-California household portfolios. The right three portfolios are defined analogously, but just focus on non-S&P 500 stocks held by the California households. Standard errors reported are Newey-West standard errors with autocorrelation up to three lags.

	Exclude CA Households, Non-S&P 500 Stocks			CA Households, Non-S&P 500 Stocks		
	Local	Nonlocal	Difference	Local	Nonlocal	Difference
Market	1.11*** (0.05)	1.16*** (0.06)	-0.05 (0.05)	1.54*** (0.18)	1.12*** (0.07)	0.42*** (0.15)
Small minus big	0.79*** (0.06)	0.82*** (0.08)	-0.04 (0.06)	0.98*** (0.24)	0.87*** (0.08)	0.11 (0.19)
High minus low	-0.04 (0.07)	-0.11** (0.05)	0.07 (0.07)	-0.23 (0.19)	-0.17** (0.05)	-0.06 (0.18)
Up minus down	-0.02 (0.07)	0.03 (0.05)	-0.05 (0.06)	-0.08 (0.19)	0.02 (0.05)	-0.09 (0.15)
p-value of regression	0.000***	0.000***	0.000***	0.317	0.000***	0.000***
Alpha	0.10 (0.18)	-0.25 (0.17)	0.35* (0.20)	-0.30 (0.49)	-0.29 (0.19)	-0.01 (0.43)
Raw return	1.67*** (0.47)	1.39*** (0.49)	0.28* (0.16)	1.59* (0.75)	1.27** (0.51)	0.32 (0.40)

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table XII: California robustness in aggregate portfolios.

6.8 Table XIII: Relative local ownership and following locals

Read Table XIII:

- The table separates nonlocal non-S&P 500 stocks into high-RLO and low-RLO groups.
- Local non-S&P 500 stocks do not significantly outperform nonlocal non-S&P 500 stocks with high RLO.
- High-RLO nonlocal non-S&P 500 stocks outperform low-RLO nonlocal non-S&P 500 stocks.
- The risk-adjusted high-RLO minus low-RLO difference is 21 basis points per month.

Economic message: The behavior of local investors is informative even to outsiders. Nonlocal investors can reduce the local-information disadvantage by favoring stocks that are disproportionately held by nearby households.

Table XIII
Relation between Household Portfolio Raw Returns and Locality Revisited (Relation between Returns and Nonlocal Share Interacted with *RLO*)

The table presents the results of fitting cross-sectional regressions of household portfolio returns on portfolio locality shares (computed at the end of 1991) with a particular emphasis on relative local ownership (*RLO*), as defined in equation (3). The sample consists of 34,517 households with at least \$1,000 in stock holdings at the end of 1991. A stock is local for a given household if the firm is headquartered within 250 miles of the household. The median *RLO* cutoff is 9.8%. On average, non-S&P 500 nonlocal stocks constitute 26.4% of the value of a household's portfolio (13.4% has high *RLO*, and 13.0% has low *RLO*). Household raw returns (calculated based on monthly rebalancing of their portfolios) are calculated over the next year, next 3 years, and next 5 years and are expressed in percentages and average excess monthly household returns α (calculated from a four-factor model according to equation (6)) are expressed in basis points. These returns are then related to the portfolio shares, computed at the end of 1991, according to the specifications outlined in equation (8). Standard errors (shown in parentheses) allow for heteroskedasticity.

	Raw Returns			Average Excess Monthly Returns
	1 Year	3 Years	5 Years	
Share of S&P 500 local stocks	10.8*** (0.5)	21.9*** (1.4)	100.8*** (3.9)	0.11*** (0.02)
Share of S&P 500 nonlocal stocks	10.0*** (0.3)	21.7*** (0.8)	99.2*** (2.4)	0.14*** (0.01)
Share of non-S&P 500 local stocks	17.8*** (1.1)	33.5*** (2.9)	128.1*** (8.4)	0.26*** (0.03)
Share of non-S&P 500 nonlocal stocks with high relative local ownership (high <i>RLO</i>)	15.7*** (1.0)	28.3*** (2.8)	115.7*** (10.4)	0.15*** (0.03)
Share of non-S&P 500 nonlocal stocks with low relative local ownership (low <i>RLO</i>)	7.3*** (1.2)	18.8*** (3.0)	73.7*** (11.0)	-0.07* (0.04)
Non-S&P 500 local – non-S&P 500 nonlocal (high <i>RLO</i>)	2.0 (1.6)	5.2 (4.1)	12.4 (13.8)	0.11** (0.05)
Non-S&P 500 nonlocal (high <i>RLO</i>) – non-S&P 500 nonlocal (low <i>RLO</i>)	8.4*** (1.7)	9.5*** (4.3)	42.9*** (16.3)	0.21*** (0.05)
<i>p</i> -value: All non-S&P 500 coefficients the same	0.000***	0.003***	0.001***	0.000***
R^2	0.0049	0.0030	0.0054	0.0028
Number of observations	27,032	12,067	6,462	26,093

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

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Table XIII: Household portfolio returns and nonlocal shares interacted with relative local ownership.

7 Short summary

- Households have a strong local bias: they allocate about one-third of their stock portfolios to firms headquartered within 250 miles.
- The local bias is much larger than what would be implied by the local availability of firms.
- Local holdings outperform nonlocal holdings, and the return premium is concentrated in non-S&P 500 stocks.
- Risk adjustment with the market, size, value, and momentum factors does not eliminate the premium.
- The result survives controls for industry composition, geography, California, retirement-account ownership, and region.
- Aggregate zero-cost portfolios based on local-minus-nonlocal holdings earn positive abnormal returns, especially in non-S&P 500 stocks.
- Households that invest more locally are the households whose local-minus-nonlocal positions contain the strongest return signal.
- Relative local ownership provides an outside signal: nonlocal stocks heavily owned by local investors perform better than nonlocal stocks not heavily owned by locals.

8 Conclusion

8.1 Core conclusion

The paper concludes that individual investors' local bias contains value-relevant information. Households do not merely buy familiar nearby firms; their local positions perform better than their nonlocal positions, especially where information asymmetry is likely to be high.

8.2 Economic intuition

Local investors may observe local firms more closely through local news, local employment networks, suppliers, customers, community knowledge, and informal information channels. These advantages matter most for smaller, less widely followed firms, where public information is thinner.

8.3 Incremental contribution revisited

The paper extends the local-information literature from professional money managers to households. It shows that even ordinary retail investors may extract information from geography, and that the information is visible in the cross-section of their holdings.

8.4 Implications

- Local bias is not automatically irrational. It can reflect informed specialization.
- Diversification and information acquisition involve a trade-off. Households may sacrifice diversification but gain local informational advantage.
- Local ownership can be useful to outsiders as a signal of local private information.
- The informational advantage appears concentrated in less visible firms rather than blue-chip S&P 500 firms.

8.5 Limitations

The evidence is not a direct observation of information. It is based on return patterns that are consistent with an information channel and robust to many alternative explanations. The brokerage sample also may not represent all individual investors, and the results come from a specific period and institutional setting.

8.6 Takeaway

The memorable result is: **local does as local is**. Individual investors tilt toward local firms, and those local positions appear to contain useful information. The effect is strongest when local knowledge is most likely to matter: nearby, non-S&P 500 firms with less public information.